

Math 221D Quiz 1 (September 4, 2026)

1. Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, and suppose that $\mathbf{x}, \mathbf{y} \in \text{Span}(\mathbf{u}, \mathbf{v})$. Prove that $\mathbf{x} - 5\mathbf{y} \in \text{Span}(\mathbf{u}, \mathbf{v})$ as well.

Solution. The assumptions $\mathbf{x}, \mathbf{y} \in \text{Span}(\mathbf{u}, \mathbf{v})$ mean that there exist real numbers $a, b, c, d \in \mathbb{R}$ such that

$$\mathbf{x} = a\mathbf{u} + b\mathbf{v}, \quad \mathbf{y} = c\mathbf{u} + d\mathbf{v}.$$

Then

$$\mathbf{x} - 5\mathbf{y} = (a\mathbf{u} + b\mathbf{v}) - 5(c\mathbf{u} + d\mathbf{v}) = (a - 5c)\mathbf{u} + (b - 5d)\mathbf{v},$$

which shows that $\mathbf{x} - 5\mathbf{y}$ is a linear combination of $\mathbf{u}$ and $\mathbf{v}$, as desired.

2. Prove that every vector $\mathbf{x} \in \mathbb{R}^3$ which satisfies $x_1 - 2x_2 + x_3 = 0$ is a linear combination of $\mathbf{u} = (1, 0, -1)$ and $\mathbf{v} = (1, 1, 1)$.

Solution. For every $\mathbf{x}$ which satisfies the given equation, we have $x_1 = 2x_2 - x_3$, so

$$\mathbf{x} = (2x_2 - x_3, x_2, x_3) = x_2(2, 1, 0) + x_3(-1, 0, 1).$$

Now we note that $(2, 1, 0) = \mathbf{u} + \mathbf{v}$, and $(-1, 0, 1) = -\mathbf{u}$, so continuing from above,

$$\mathbf{x} = x_2(\mathbf{u} + \mathbf{v}) - x_3\mathbf{u} = (x_2 - x_3)\mathbf{u} + x_2\mathbf{v},$$

which is a linear combination of $\mathbf{u}$ and $\mathbf{v}$.